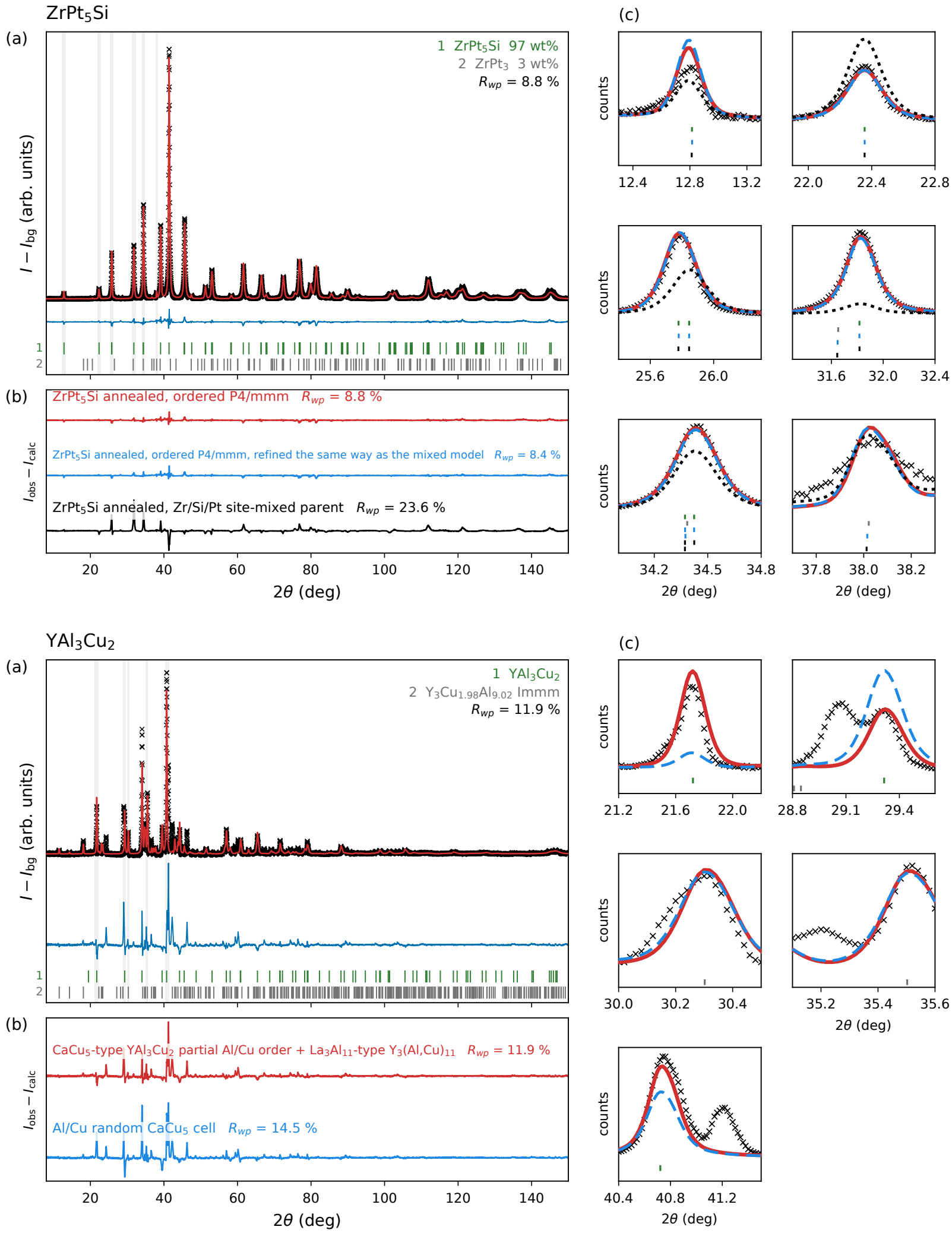
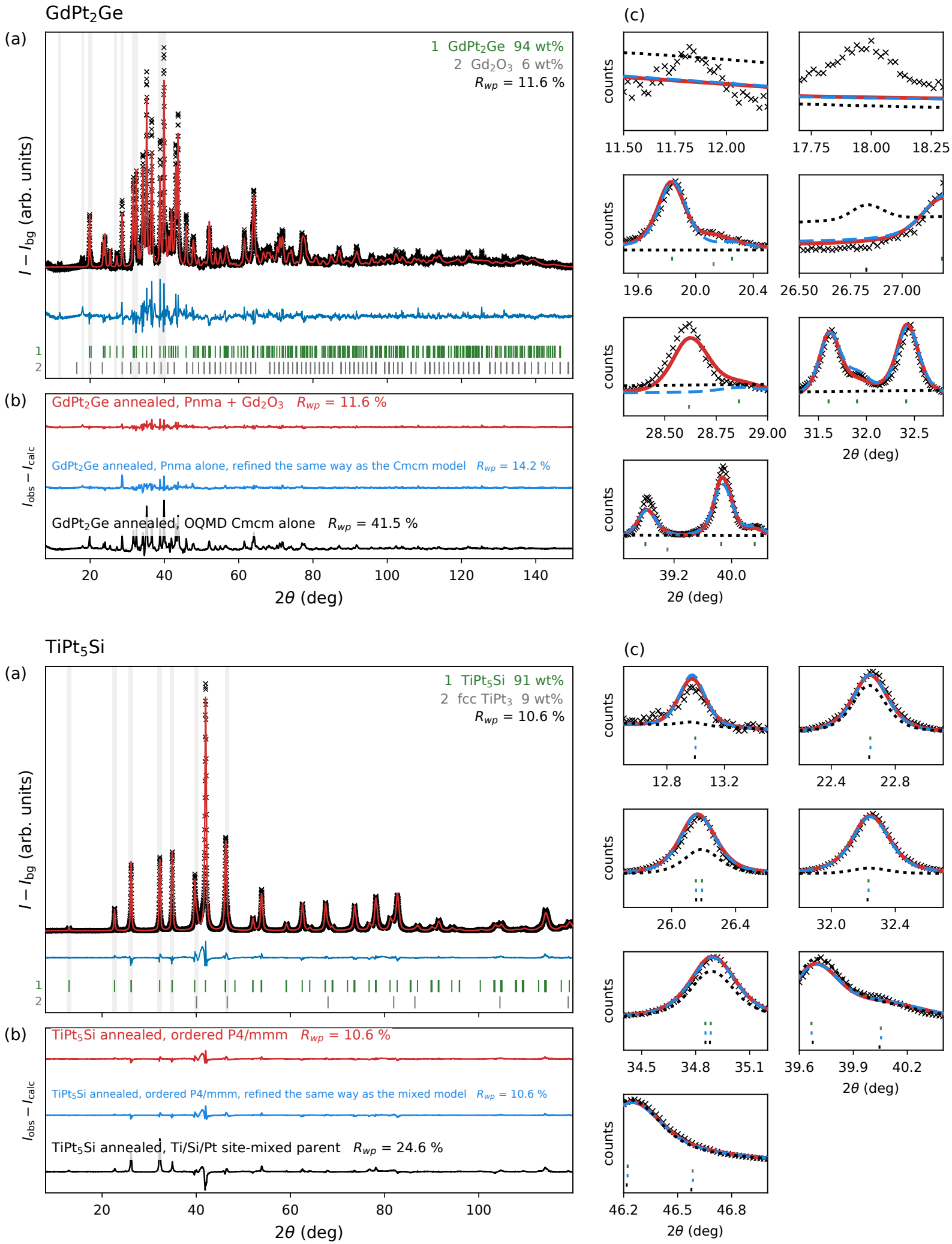
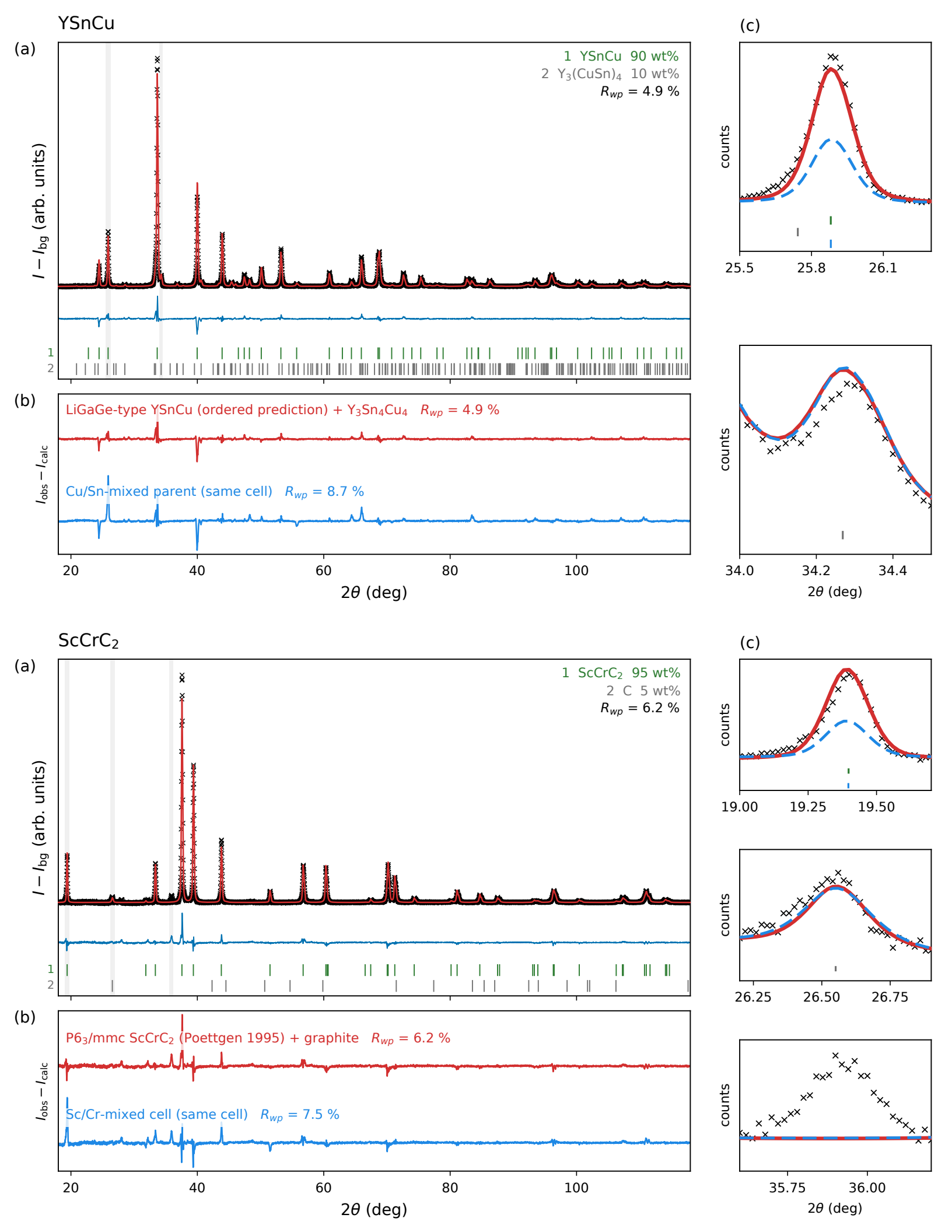
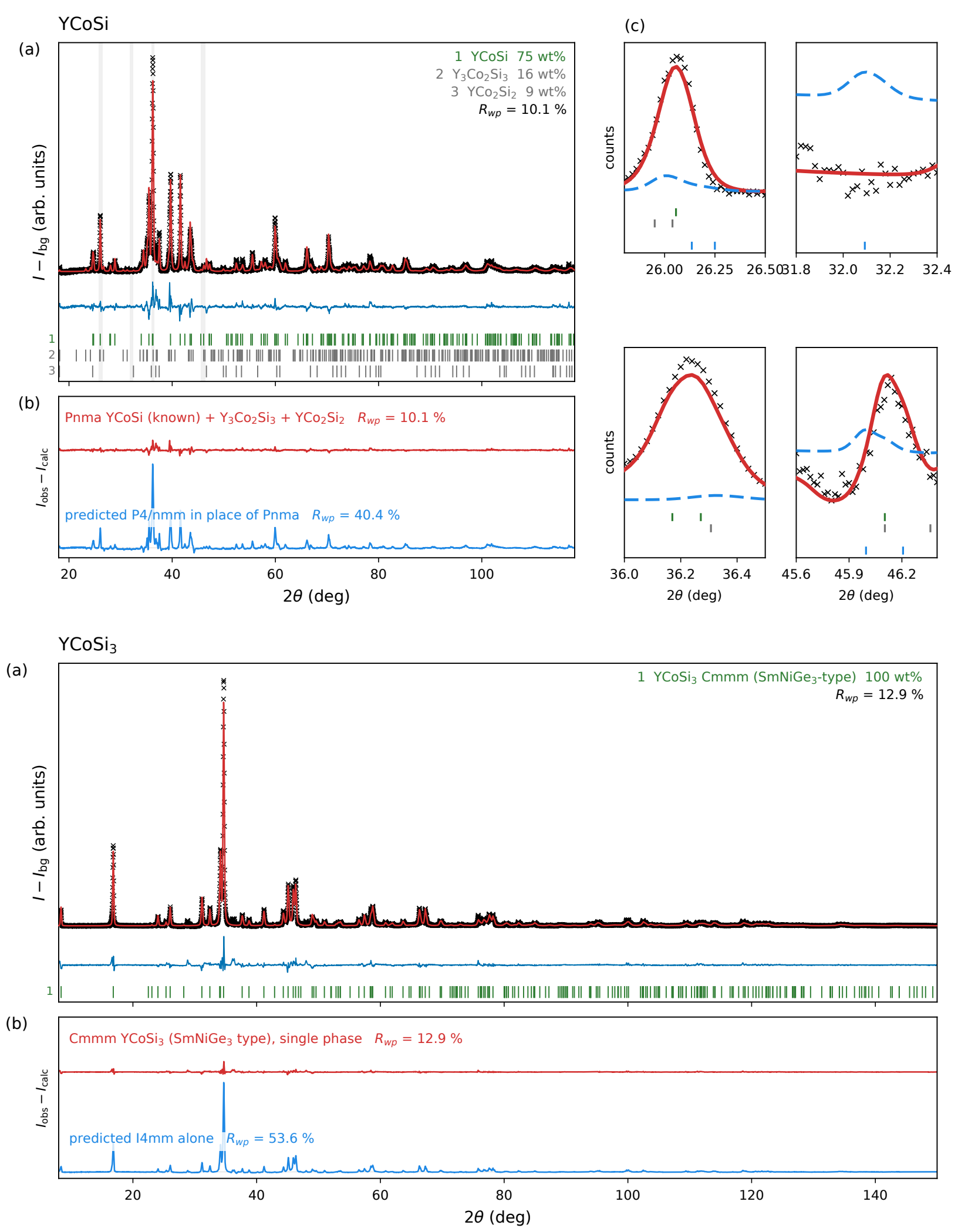
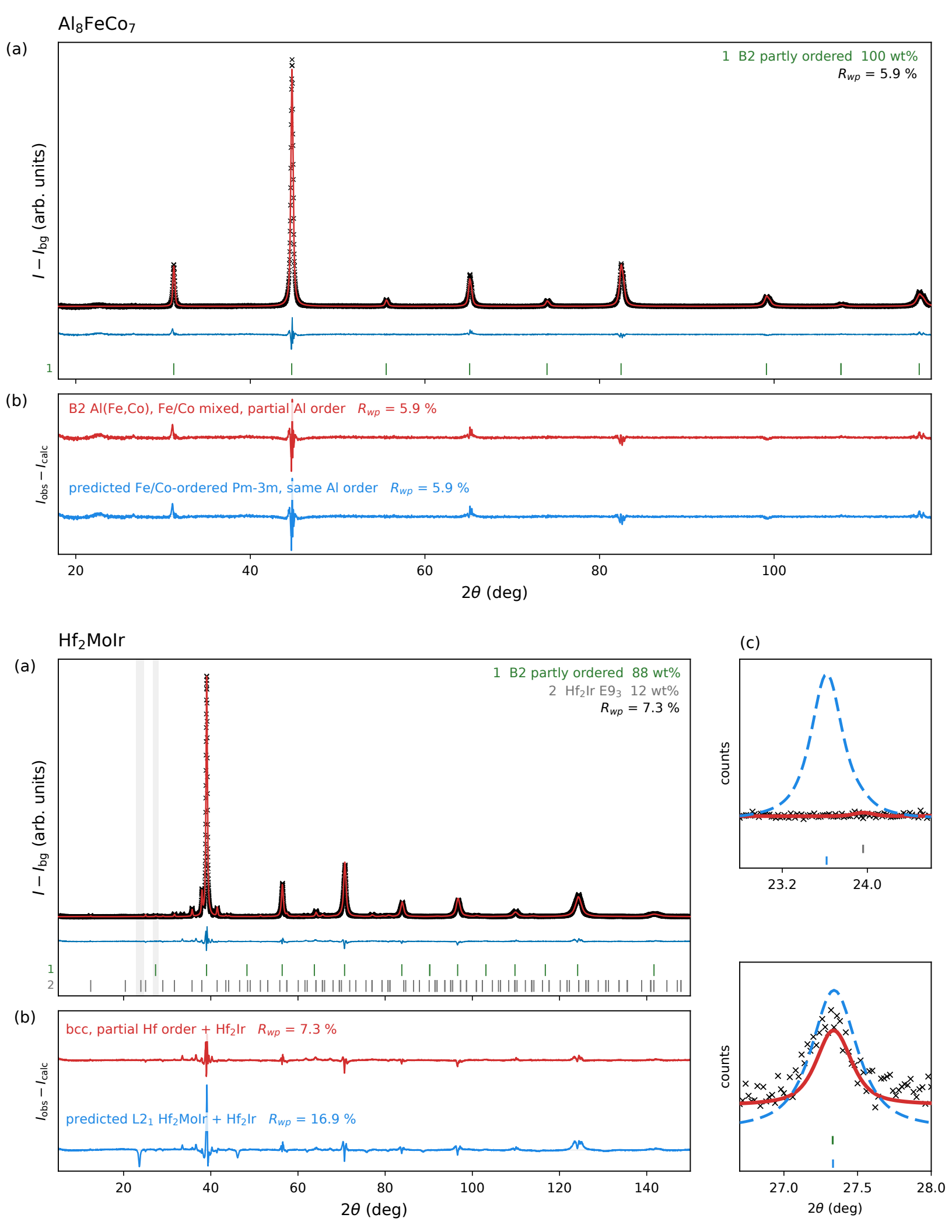
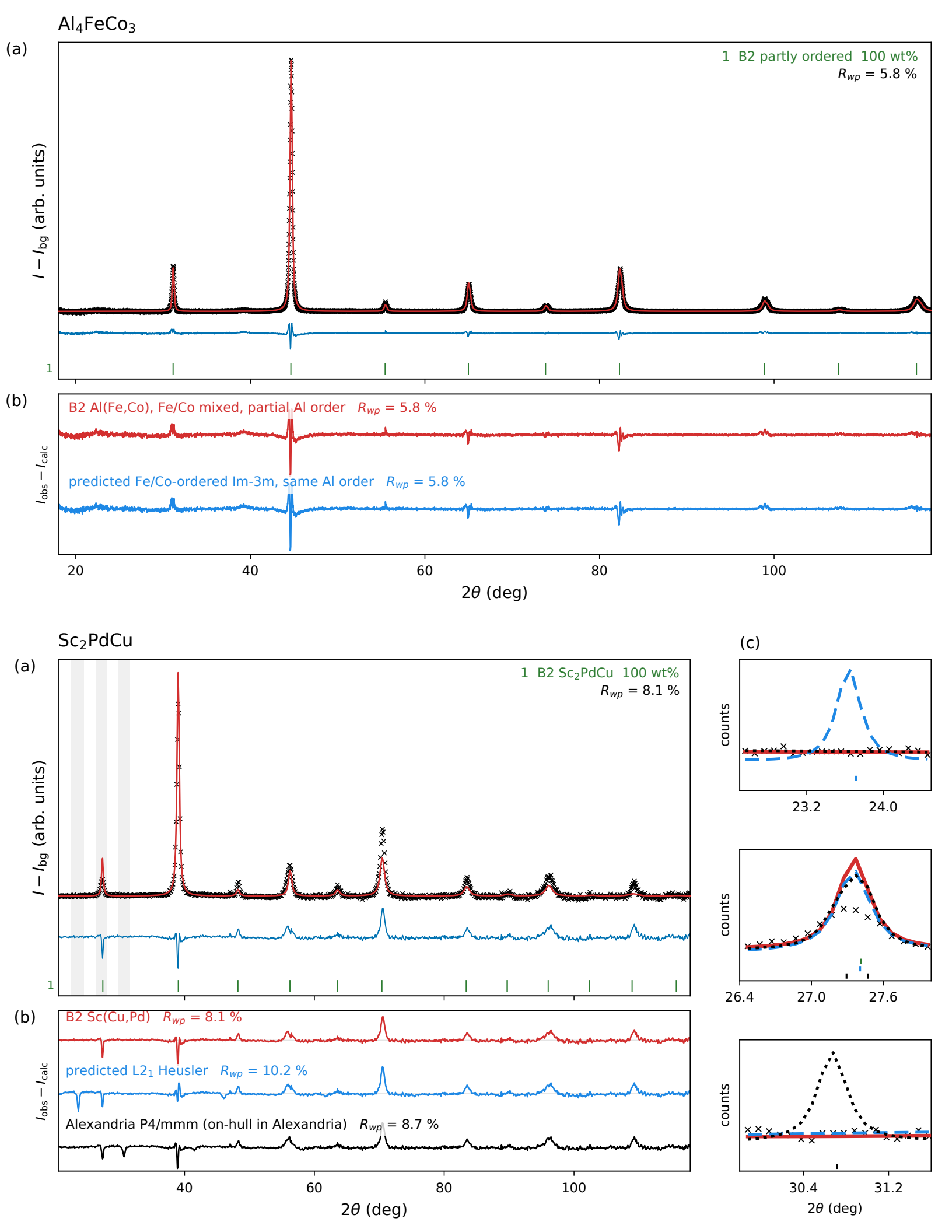

ZrPt5Si
(a)
1 ZrPt5Si 97 wt%
2 ZrPt3 3 wt%
Rwp = 8.8 %
I − Ibg (arb. units)
(b)
ZrPt5Si annealed, ordered P4/mmm Rwp = 8.8 %
ZrPt5Si annealed, ordered P4/mmm, refined the same way as the mixed model Rwp = 8.4 %
ZrPt5Si annealed, Zr/Si/Pt site-mixed parent Rwp = 23.6 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)
YAl3Cu2
(a)
1 YAl3Cu2
2 Y3Cu1.98Al9.02 Immm
Rwp = 11.9 %
I − Ibg (arb. units)
(b)
CaCu5-type YAl3Cu2 partial Al/Cu order + La3Al11-type Y3(Al,Cu)11 Rwp = 11.9 %
Al/Cu random CaCu5 cell Rwp = 14.5 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

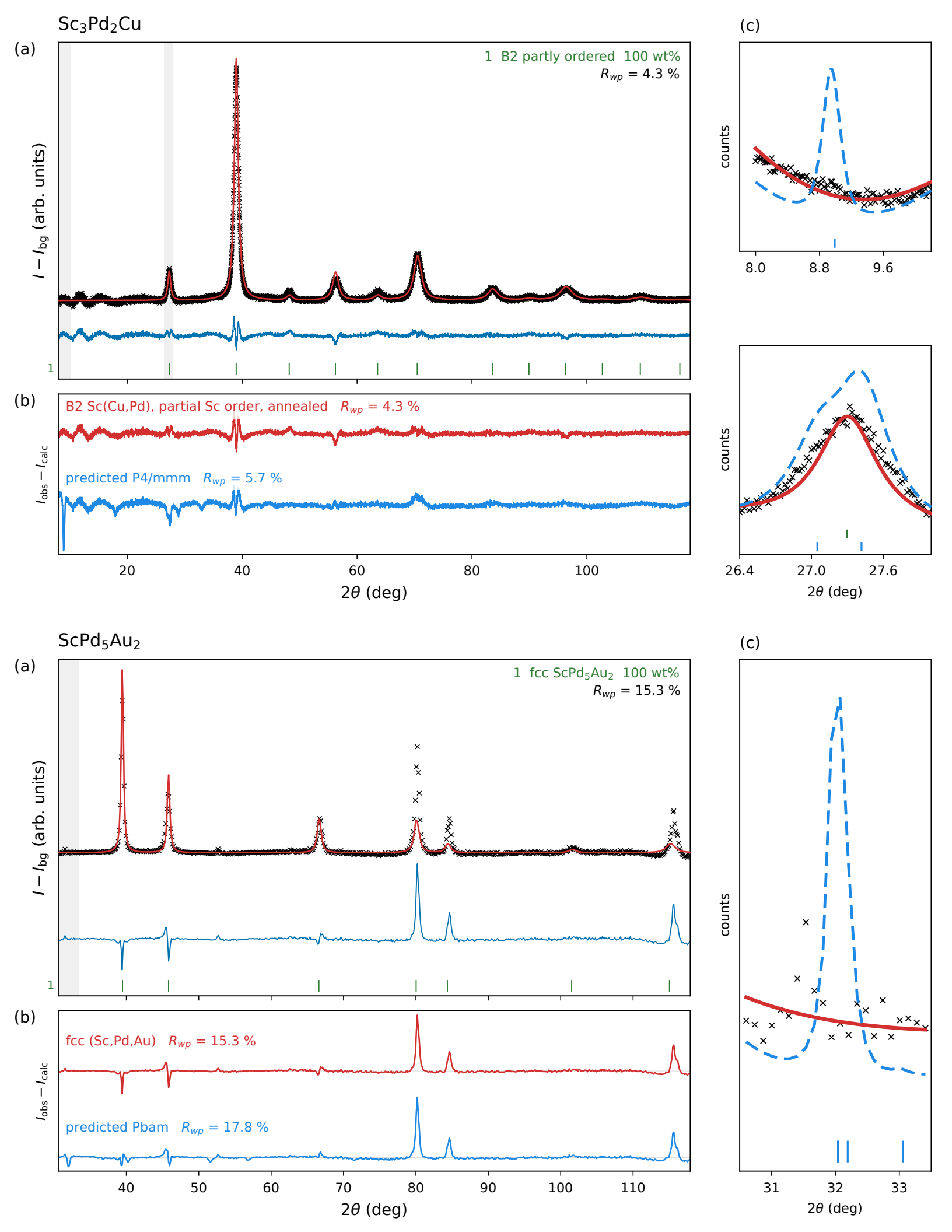

GdPt2Ge
(a)
1 GdPt2Ge 94 wt%
2 Gd2O3 6 wt%
Rwp = 11.6 %
I − Ibg (arb. units)
(b)
GdPt2Ge annealed, Pnma + Gd2O3 Rwp = 11.6 %
GdPt2Ge annealed, Pnma alone, refined the same way as the Cmcm model Rwp = 14.2 %
GdPt2Ge annealed, OQMD Cmcm alone Rwp = 41.5 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)
TiPt5Si
(a)
1 TiPt5Si 91 wt%
2 fcc TiPt3 9 wt%
Rwp = 10.6 %
I − Ibg (arb. units)
(b)
TiPt5Si annealed, ordered P4/mmm Rwp = 10.6 %
TiPt5Si annealed, ordered P4/mmm, refined the same way as the mixed model Rwp = 10.6 %
TiPt5Si annealed, Ti/Si/Pt site-mixed parent Rwp = 24.6 %
Iobs − Icalc
2θ (deg)
(c)
counts
2θ (deg)

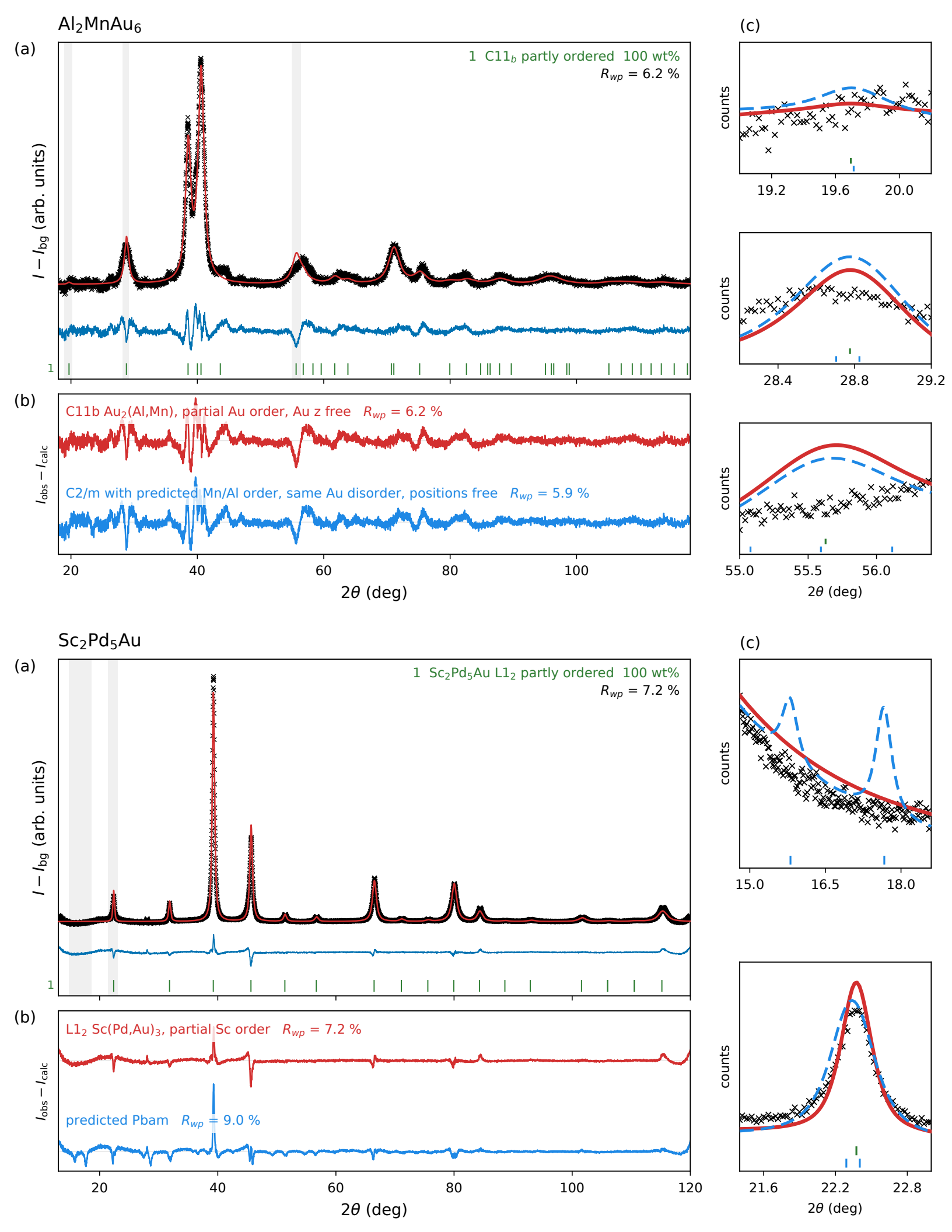

YSnCu
(a)
1 YSnCu 90 wt%
2 Y3(CuSn)4 10 wt%
Rwp = 4.9 %
I − Ibg (arb. units)
1
2
(b)
LiGaGe-type YSnCu (ordered prediction) + Y3Sn4Cu4 Rwp = 4.9 %
Cu/Sn-mixed parent (same cell) Rwp = 8.7 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
25.5
25.8
26.1
34.0
34.2
34.4
2θ (deg)
ScCrC2
(a)
1 ScCrC2 95 wt%
2 C 5 wt%
Rwp = 6.2 %
I − Ibg (arb. units)
1
2
(b)
P63/mmc ScCrC2 (Poettgen 1995) + graphite Rwp = 6.2 %
Sc/Cr-mixed cell (same cell) Rwp = 7.5 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
19.00
19.25
19.50
26.25
26.50
26.75
35.75
36.00
2θ (deg)

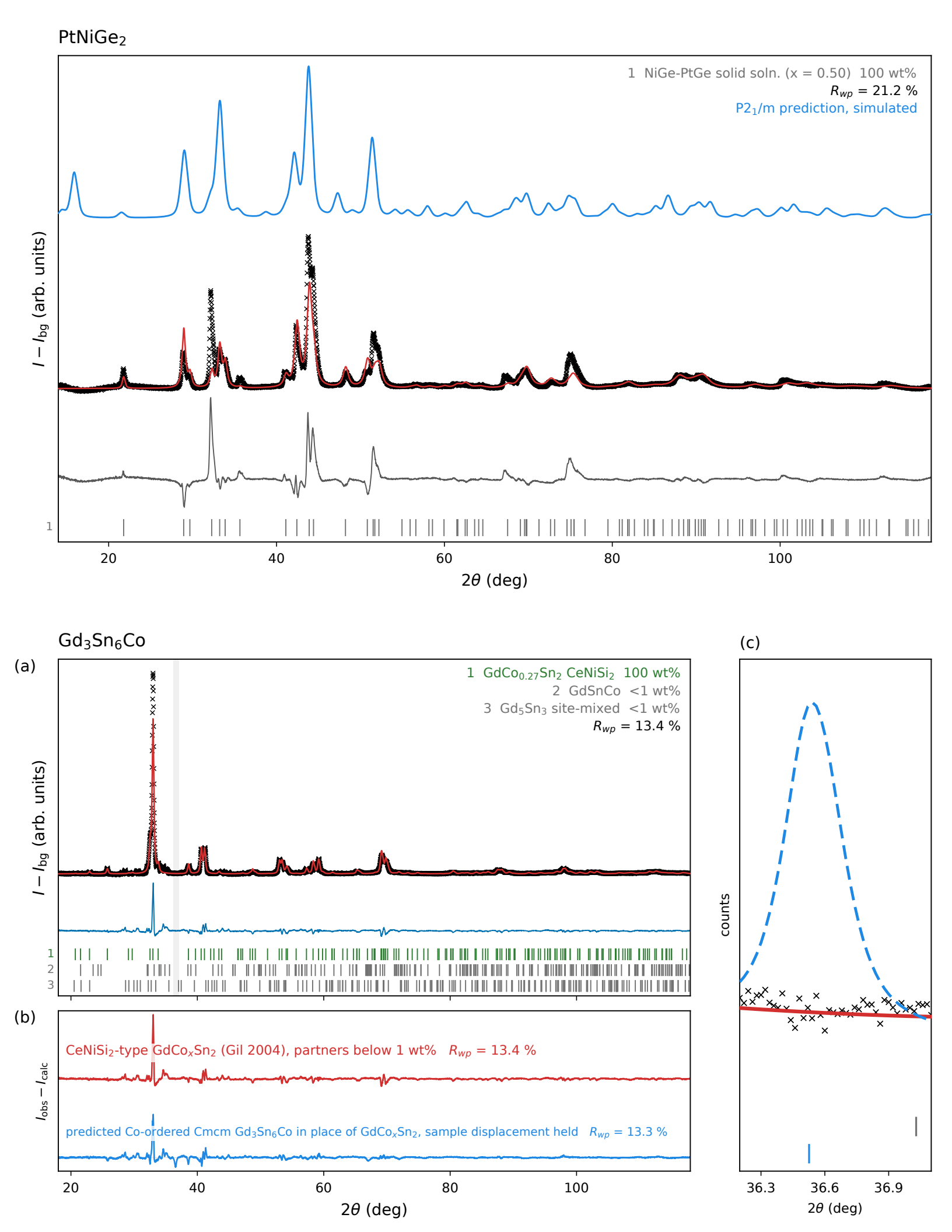

YCoSi
(a)
1 YCoSi 75 wt%
2 Y3Co2Si3 16 wt%
3 YCo2Si2 9 wt%
Rwp = 10.1 %
I − Ibg (arb. units)
(b)
Pnma YCoSi (known) + Y3Co2Si3 + YCo2Si2 Rwp = 10.1 %
predicted P4/nmm in place of Pnma Rwp = 40.4 %
Iobs − Icalc
2θ (deg)
(c)
counts
YCoSi3
1 YCoSi3 Cmmm (SmNiGe3-type) 100 wt%
Rwp = 12.9 %
Cmmm YCoSi3 (SmNiGe3 type), single phase Rwp = 12.9 %
predicted I4mm alone Rwp = 53.6 %

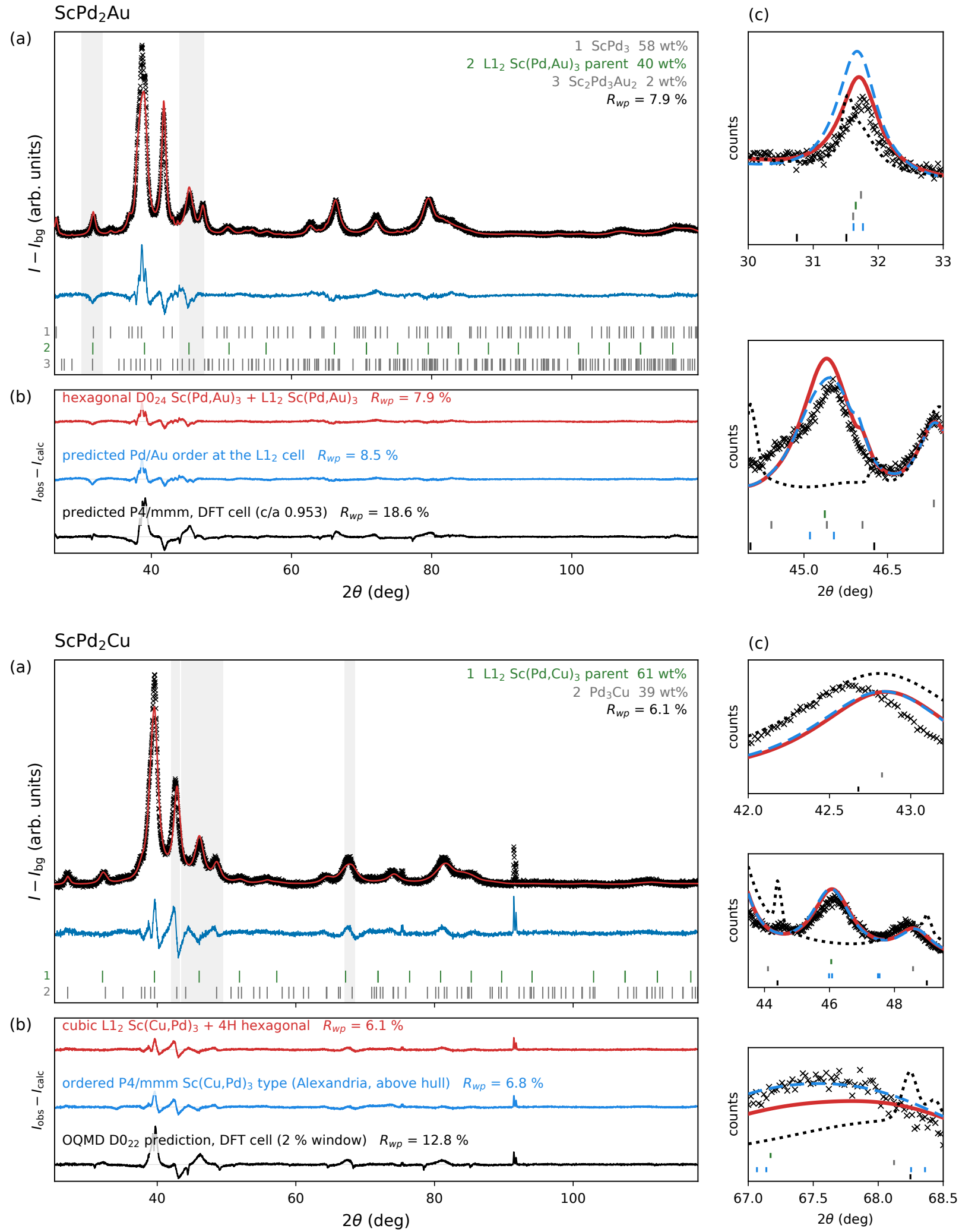

Al8FeCo7
(a)
1 B2 partly ordered 100 wt%
Rwp = 5.9 %
I − Ibg (arb. units)
1
(b)
B2 Al(Fe,Co), Fe/Co mixed, partial Al order Rwp = 5.9 %
predicted Fe/Co-ordered Pm-3m, same Al order Rwp = 5.9 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
Hf2MoIr
(a)
1 B2 partly ordered 88 wt%
2 Hf2Ir E93 12 wt%
Rwp = 7.3 %
I − Ibg (arb. units)
1
2
(b)
bcc, partial Hf order + Hf2Ir Rwp = 7.3 %
predicted L21 Hf2MoIr + Hf2Ir Rwp = 16.9 %
Iobs − Icalc
20
40
60
80
100
120
140
2θ (deg)
(c)
counts
23.2
24.0
counts
27.0
27.5
28.0
2θ (deg)

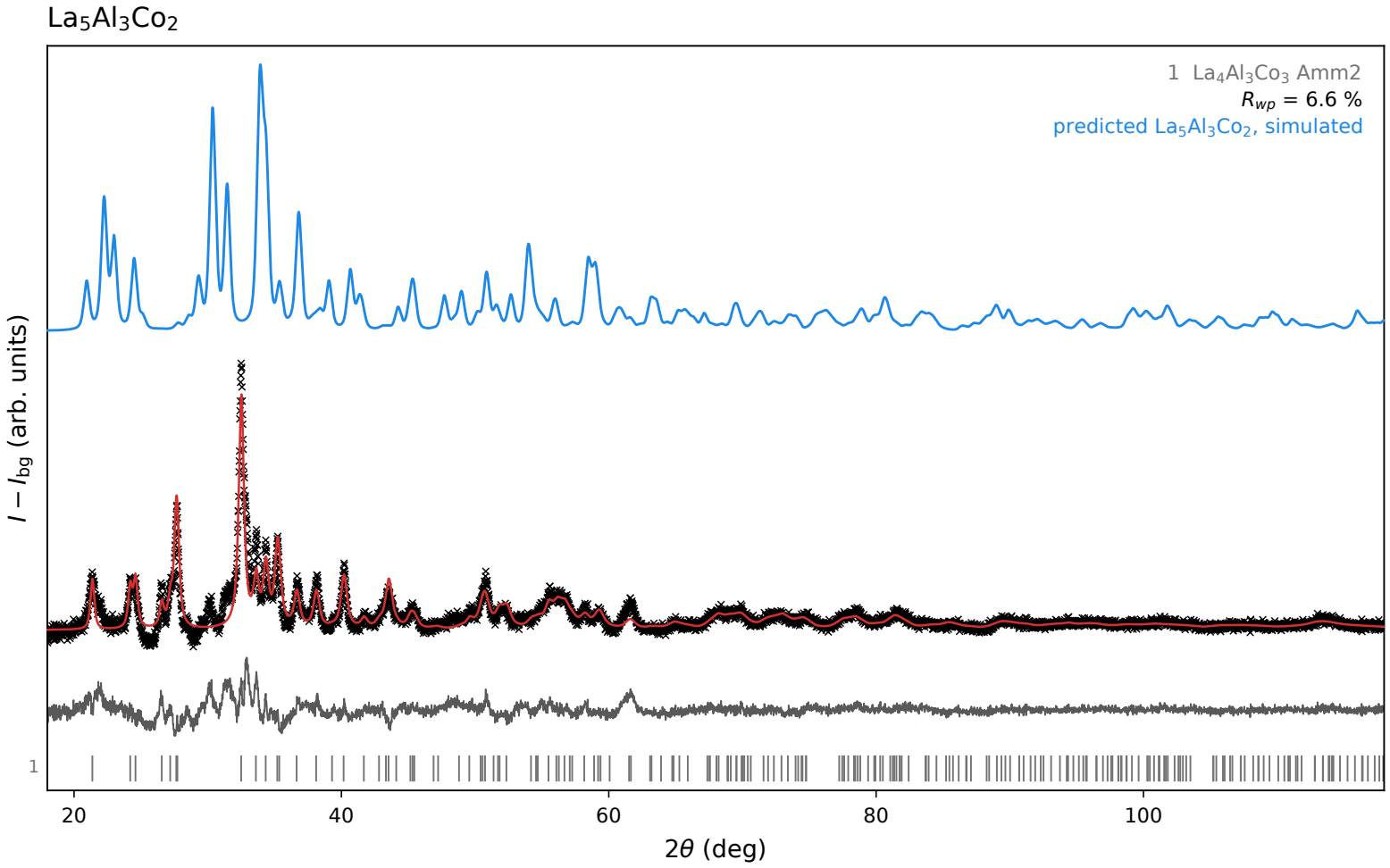

Al4FeCo3
(a)
1 B2 partly ordered 100 wt%
Rwp = 5.8 %
I − Ibg (arb. units)
1
(b)
B2 Al(Fe,Co), Fe/Co mixed, partial Al order Rwp = 5.8 %
predicted Fe/Co-ordered Im-3m, same Al order Rwp = 5.8 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
Sc2PdCu
(a)
1 B2 Sc2PdCu 100 wt%
Rwp = 8.1 %
I − Ibg (arb. units)
1
(b)
B2 Sc(Cu,Pd) Rwp = 8.1 %
predicted L21 Heusler Rwp = 10.2 %
Alexandria P4/mmm (on-hull in Alexandria) Rwp = 8.7 %
Iobs − Icalc
40
60
80
100
2θ (deg)
(c)
counts
23.2
24.0
counts
26.4
27.0
27.6
counts
30.4
31.2
2θ (deg)

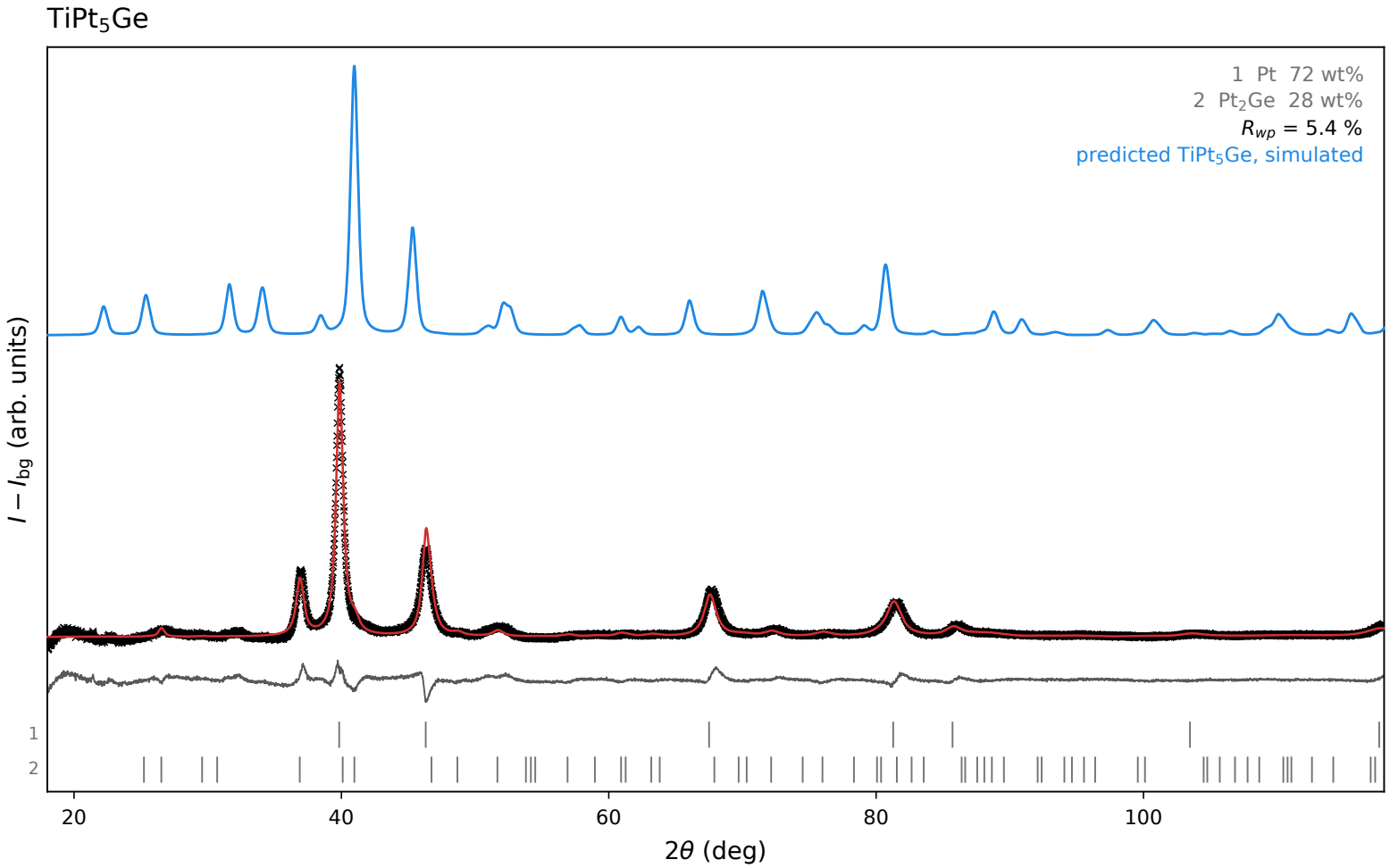

Sc3Pd2Cu
(a)
1 B2 partly ordered 100 wt%
Rwp = 4.3 %
I − Ibg (arb. units)
1
(b)
B2 Sc(Cu,Pd), partial Sc order, annealed Rwp = 4.3 %
predicted P4/mmm Rwp = 5.7 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
8.0
8.8
9.6
counts
26.4
27.0
27.6
2θ (deg)
ScPd5Au2
(a)
1 fcc ScPd5Au2 100 wt%
Rwp = 15.3 %
I − Ibg (arb. units)
1
(b)
fcc (Sc,Pd,Au) Rwp = 15.3 %
predicted Pbam Rwp = 17.8 %
Iobs − Icalc
40
50
60
70
80
90
100
110
2θ (deg)
(c)
counts
31
32
33
2θ (deg)

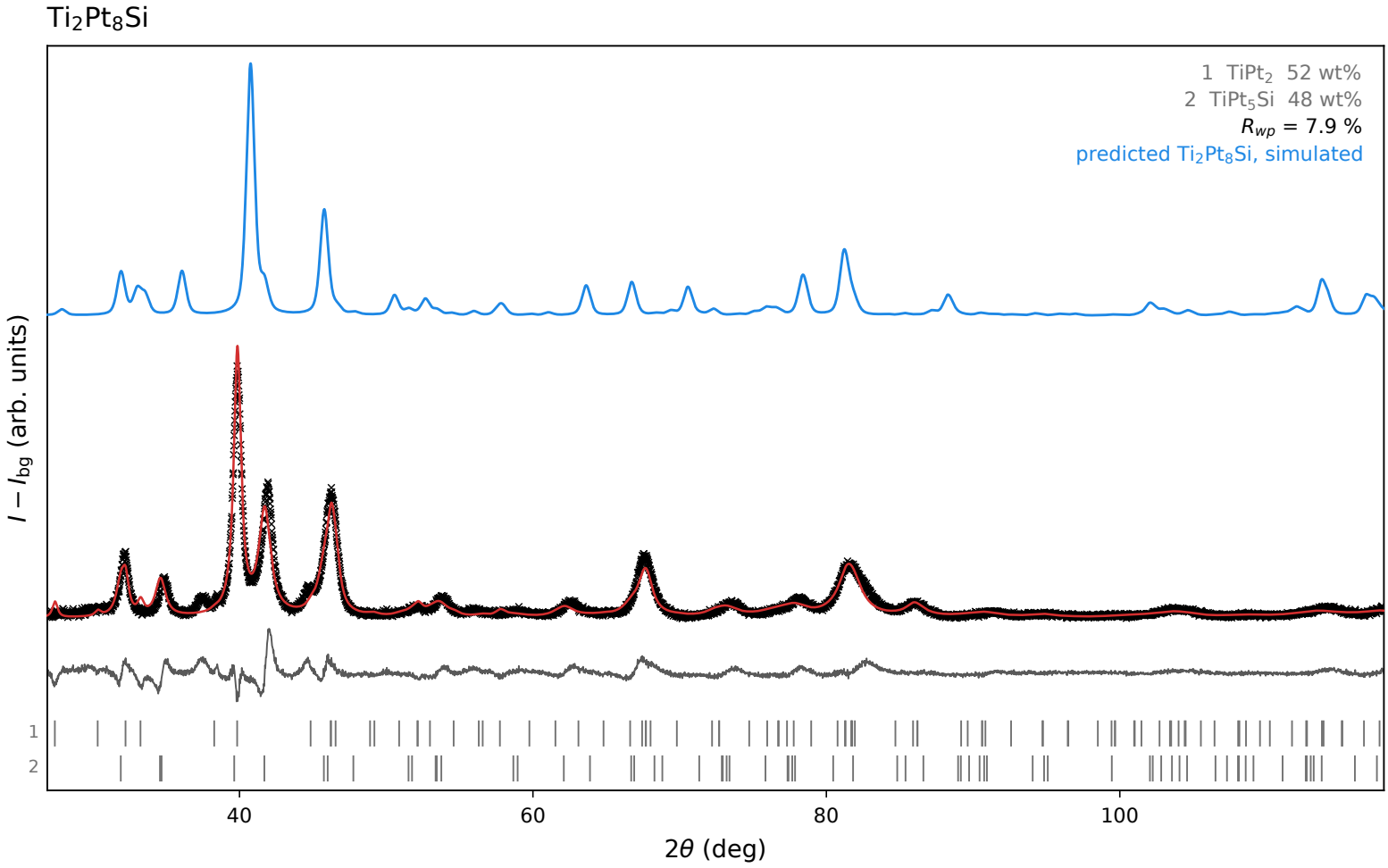

Al2MnAu6
(a)
1 C11b partly ordered 100 wt%
Rwp = 6.2 %
I − Ibg (arb. units)
1
(b)
C11b Au2(Al,Mn), partial Au order, Au z free Rwp = 6.2 %
C2/m with predicted Mn/Al order, same Au disorder, positions free Rwp = 5.9 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
19.2
19.6
20.0
28.4
28.8
29.2
55.0
55.5
56.0
2θ (deg)
Sc2Pd5Au
(a)
1 Sc2Pd5Au L12 partly ordered 100 wt%
Rwp = 7.2 %
I − Ibg (arb. units)
1
(b)
L12 Sc(Pd,Au)3, partial Sc order Rwp = 7.2 %
predicted Pbam Rwp = 9.0 %
Iobs − Icalc
20
40
60
80
100
120
2θ (deg)
(c)
counts
15.0
16.5
18.0
21.6
22.2
22.8
2θ (deg)

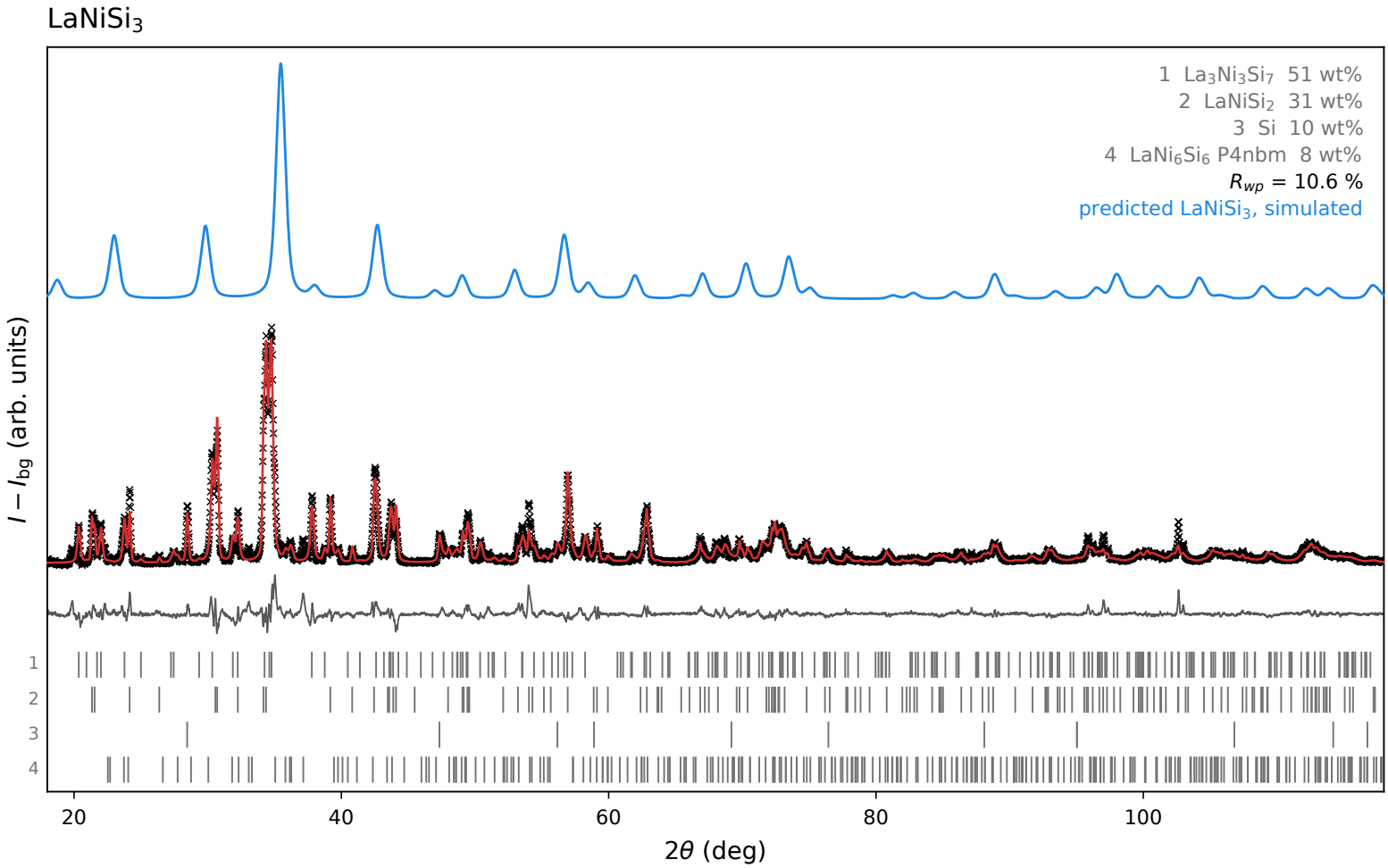

PtNiGe2
1 NiGe-PtGe solid soln. (x = 0.50) 100 wt%
Rwp = 21.2 %
P21/m prediction, simulated
I − Ibg (arb. units)
1
20
40
60
80
100
2θ (deg)
Gd3Sn6Co
(a)
1 GdCo0.27Sn2 CeNiSi2 100 wt%
2 GdSnCo <1 wt%
3 Gd5Sn3 site-mixed <1 wt%
Rwp = 13.4 %
I − Ibg (arb. units)
1
2
3
(b)
CeNiSi2-type GdCoxSn2 (Gil 2004), partners below 1 wt% Rwp = 13.4 %
predicted Co-ordered Cmcm Gd3Sn6Co in place of GdCoxSn2, sample displacement held Rwp = 13.3 %
Iobs − Icalc
20
40
60
80
100
2θ (deg)
(c)
counts
36.3
36.6
36.9
2θ (deg)

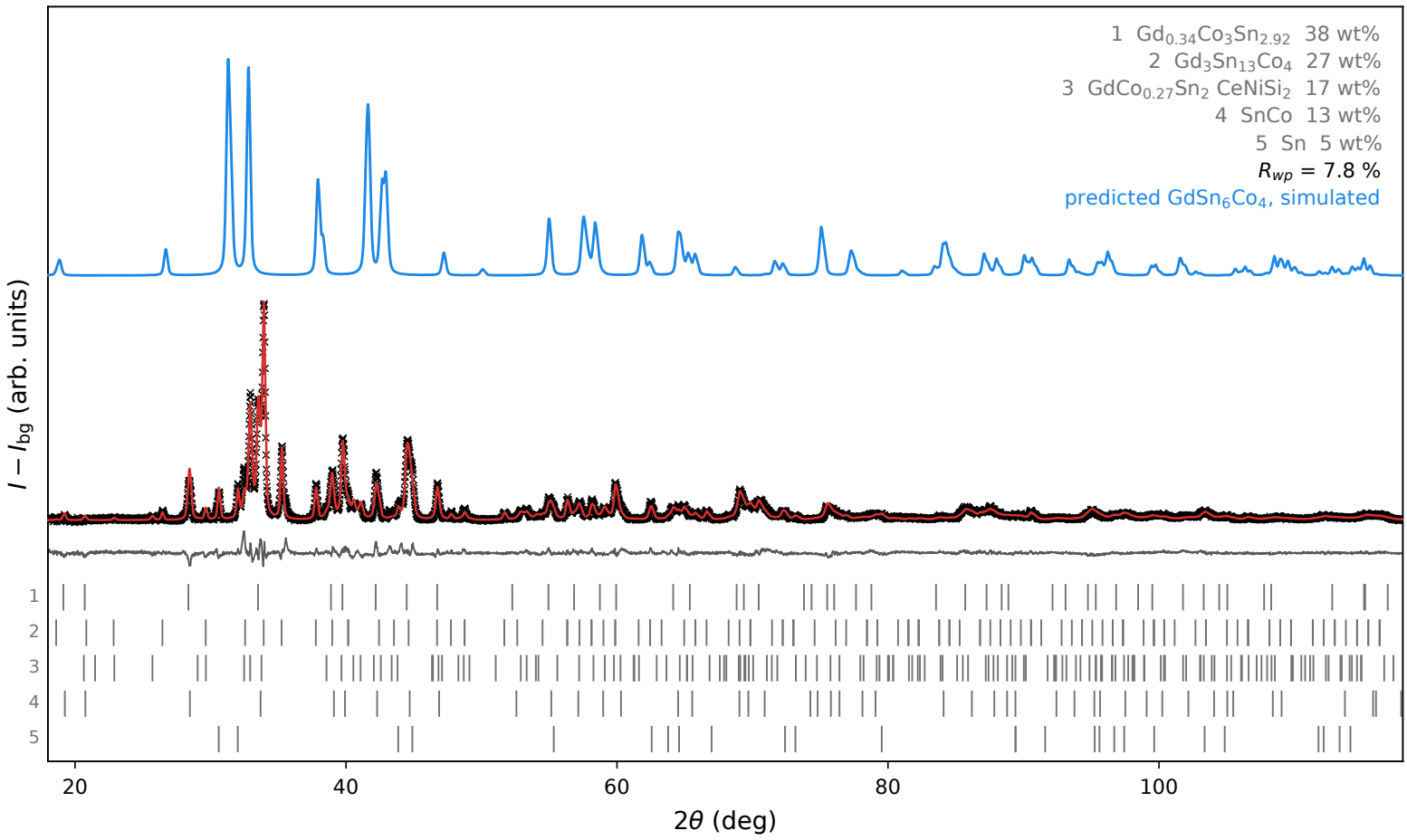

ScPd2Au
(a)
1 ScPd3 58 wt%
2 L12 Sc(Pd,Au)3 parent 40 wt%
3 Sc2Pd3Au2 2 wt%
Rwp = 7.9 %
I − Ibg (arb. units)
(b)
hexagonal D024 Sc(Pd,Au)3 + L12 Sc(Pd,Au)3 Rwp = 7.9 %
predicted Pd/Au order at the L12 cell Rwp = 8.5 %
predicted P4/mmm, DFT cell (c/a 0.953) Rwp = 18.6 %
Iobs − Icalc
2θ (deg)
(c)
counts
ScPd2Cu
(a)
1 L12 Sc(Pd,Cu)3 parent 61 wt%
2 Pd3Cu 39 wt%
Rwp = 6.1 %
(b)
cubic L12 Sc(Cu,Pd)3 + 4H hexagonal Rwp = 6.1 %
ordered P4/mmm Sc(Cu,Pd)3 type (Alexandria, above hull) Rwp = 6.8 %
OQMD D022 prediction, DFT cell (2 % window) Rwp = 12.8 %

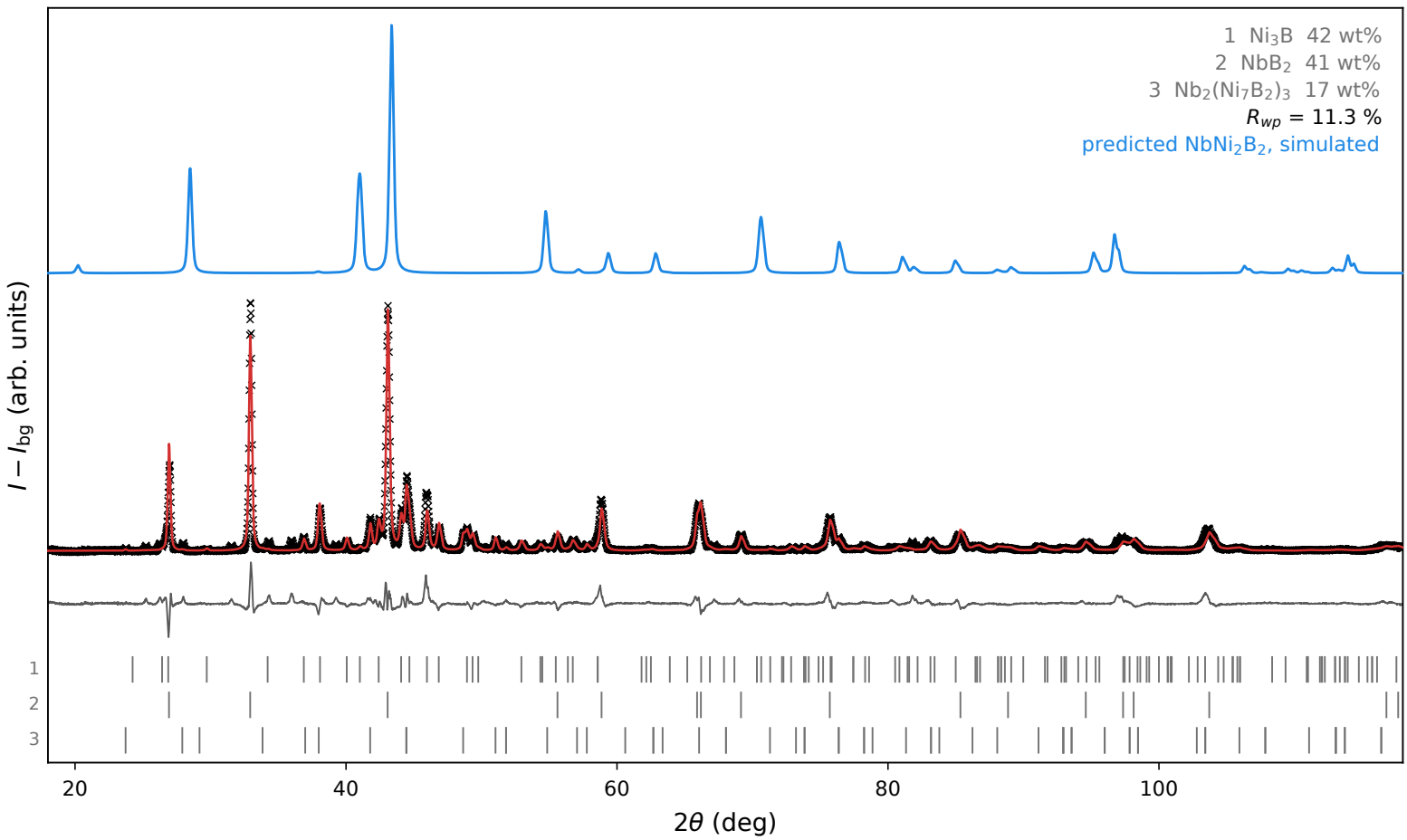

La5Al3Co2
1 La4Al3Co3 Amm2
Rwp = 6.6 %
predicted La5Al3Co2, simulated
I − Ibg (arb. units)
1
20
40
60
80
100
2θ (deg)

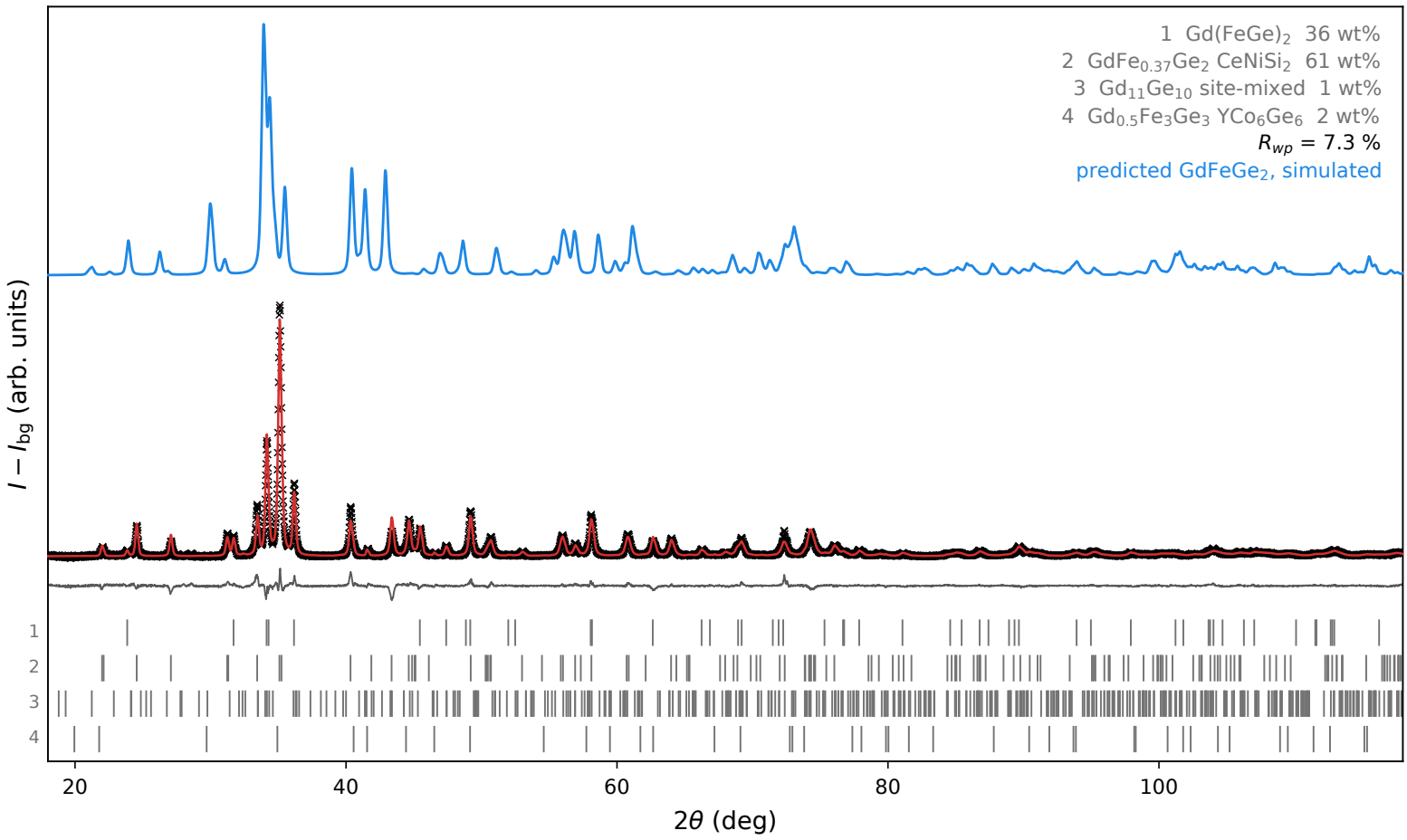

TiPt5Ge
1 Pt 72 wt%
2 Pt2Ge 28 wt%
Rwp = 5.4 %
predicted TiPt5Ge, simulated
I − Ibg (arb. units)
1
2
20
40
60
80
100
2θ (deg)

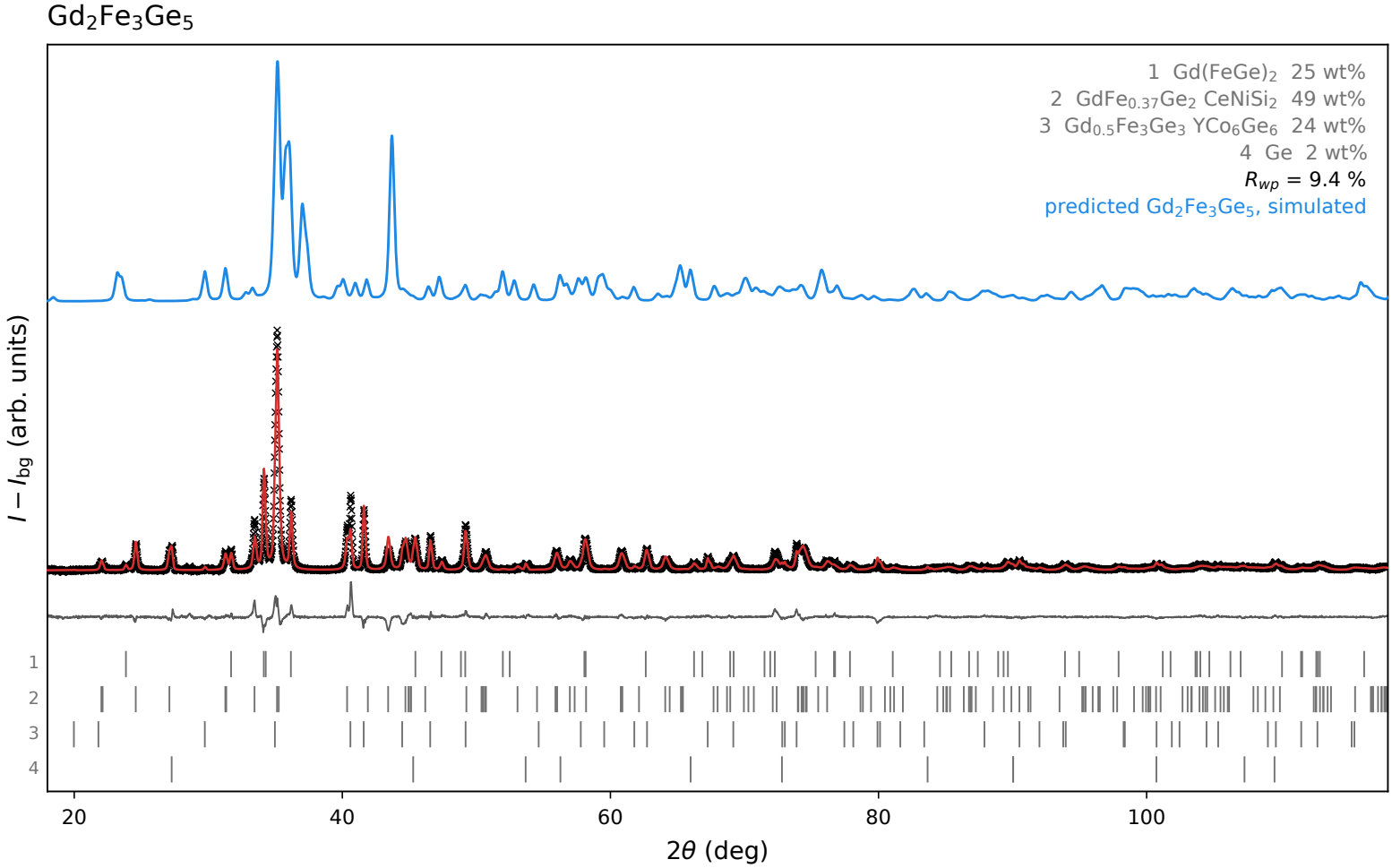

Ti2Pt8Si
1 TiPt2 52 wt%
2 TiPt5Si 48 wt%
Rwp = 7.9 %
predicted Ti2Pt8Si, simulated
I − Ibg (arb. units)
1
2
40
60
80
100
2θ (deg)

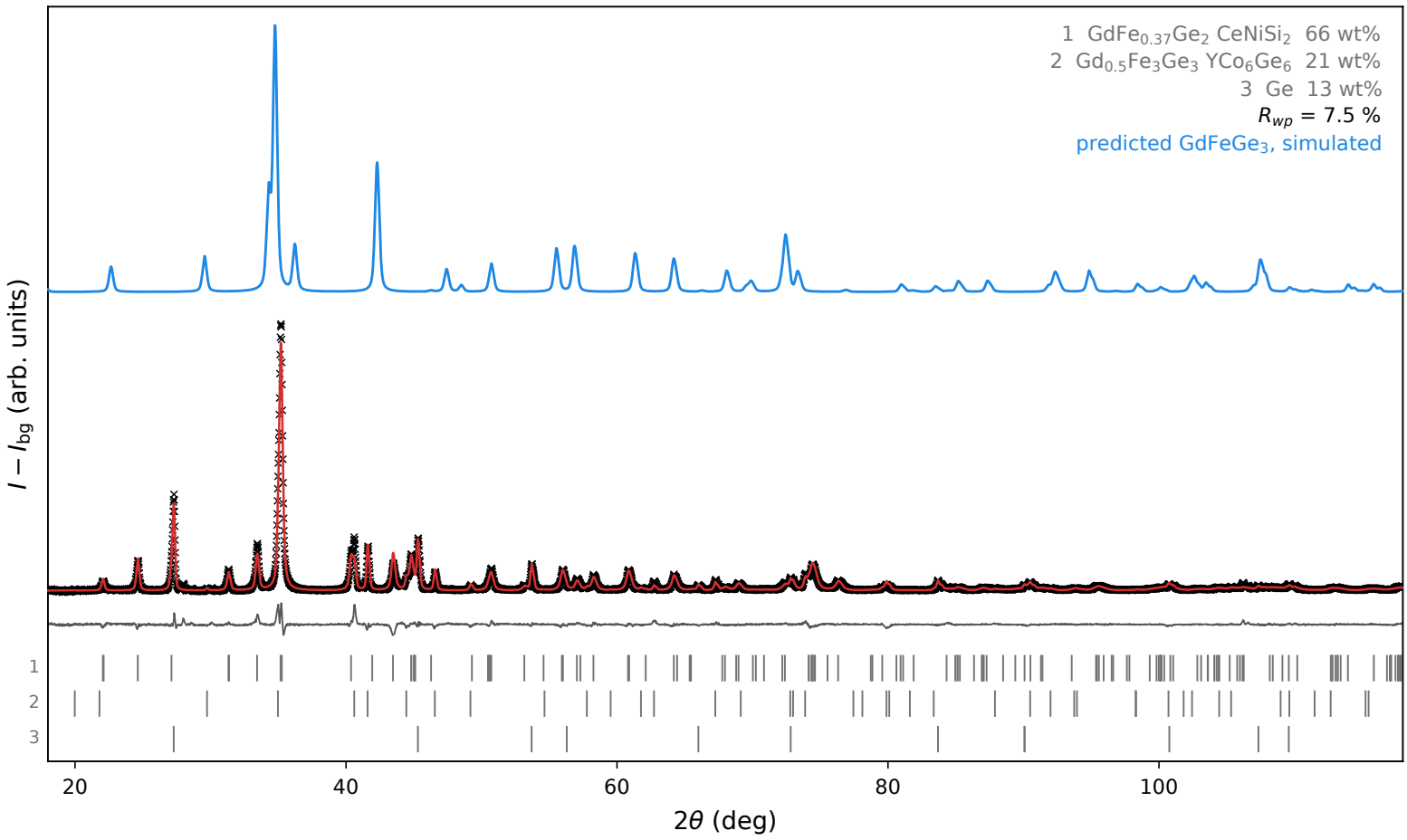

LaNiSi3
1 La3Ni3Si7 51 wt%
2 LaNiSi2 31 wt%
3 Si 10 wt%
4 LaNi6Si6 P4nbm 8 wt%
Rwp = 10.6 %
predicted LaNiSi3, simulated
I − Ibg (arb. units)
1
2
3
4
20
40
60
80
100
2θ (deg)

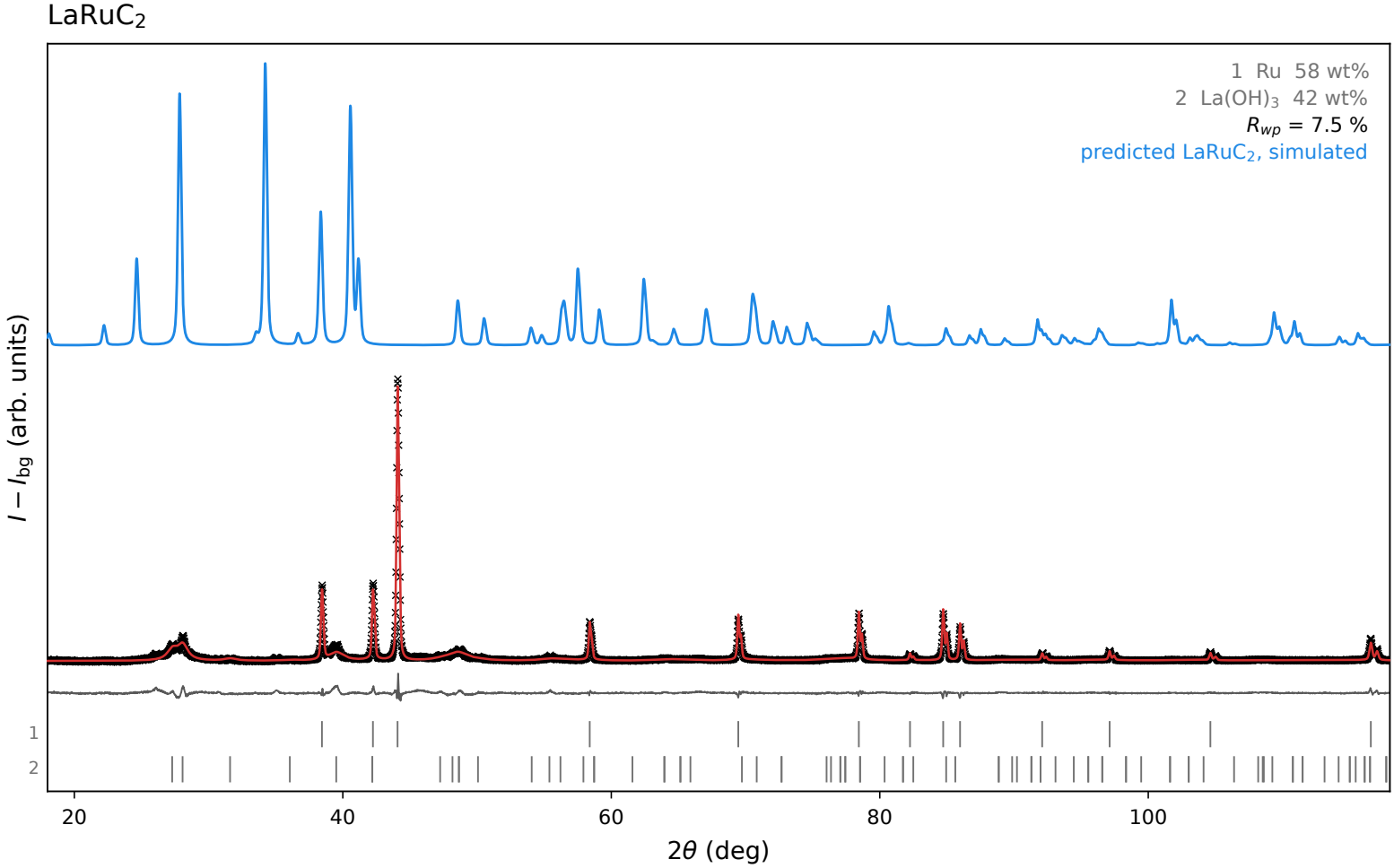

GdSn6Co4
1 Gd0.34Co3Sn2.92 38 wt%
2 Gd3Sn13Co4 27 wt%
3 GdCo0.27Sn2 CeNiSi2 17 wt%
4 SnCo 13 wt%
5 Sn 5 wt%
Rwp = 7.8 %
predicted GdSn6Co4, simulated
I − Ibg (arb. units)
1
2
3
4
5
20
40
60
80
100
2θ (deg)

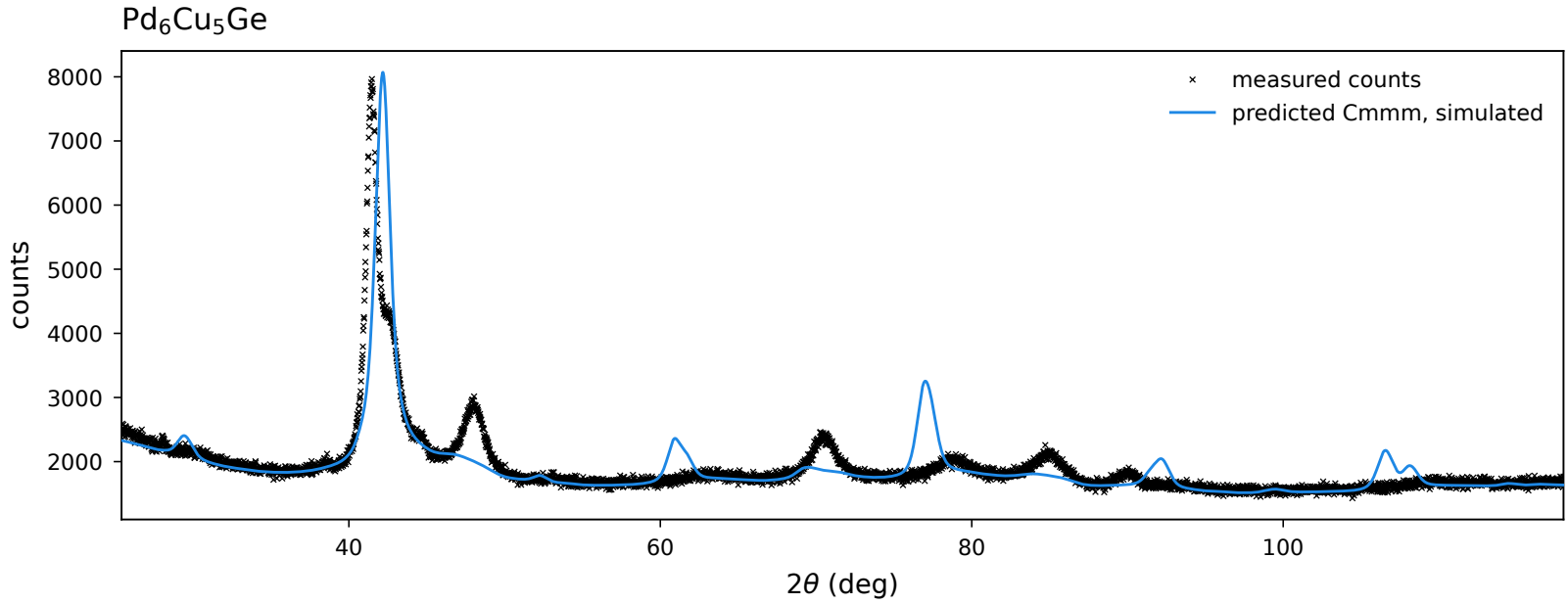

NbNi2B2
1 Ni3B 42 wt%
2 NbB2 41 wt%
3 Nb2(Ni7B2)3 17 wt%
Rwp = 11.3 %
predicted NbNi2B2, simulated
I − Ibg (arb. units)
1
2
3
20
40
60
80
100
2θ (deg)

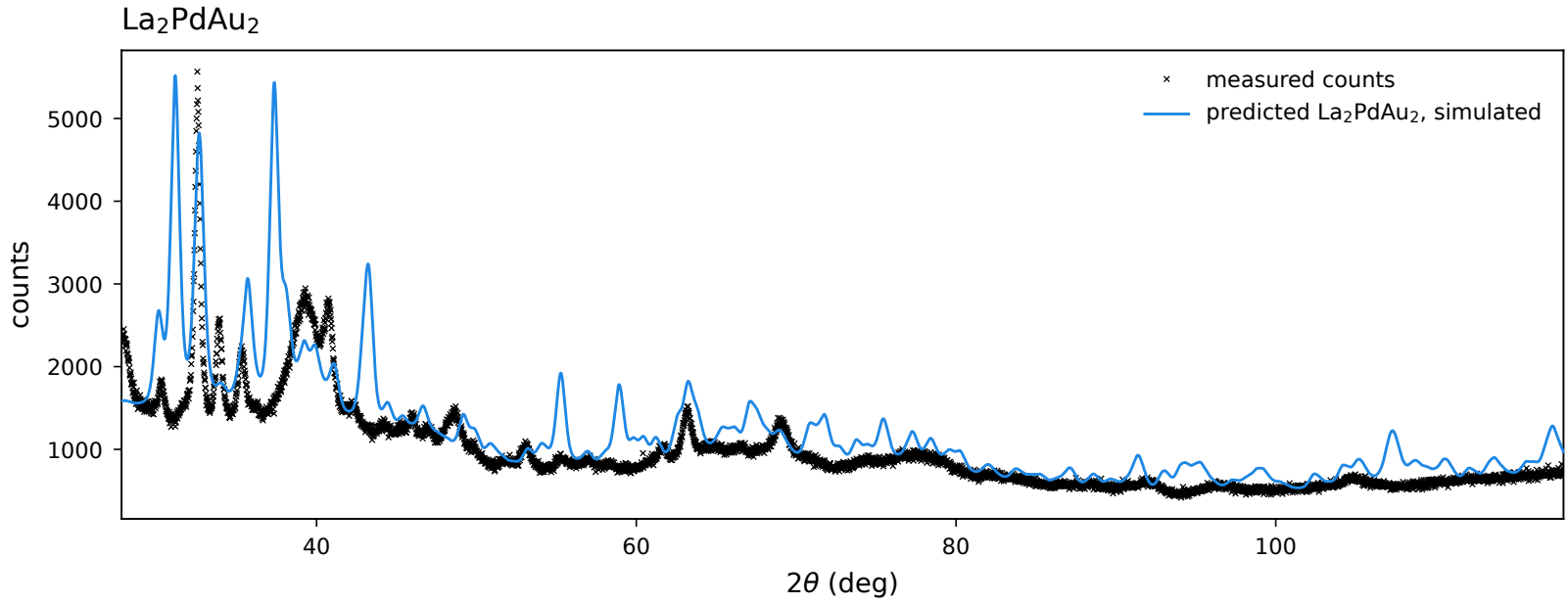

GdFeGe2
1 Gd(FeGe)2 36 wt%
2 GdFe0.37Ge2 CeNiSi2 61 wt%
3 Gd11Ge10 site-mixed 1 wt%
4 Gd0.5Fe3Ge3 YCo6Ge6 2 wt%
Rwp = 7.3 %
predicted GdFeGe2, simulated
I − Ibg (arb. units)
1
2
3
4
20
40
60
80
100
2θ (deg)

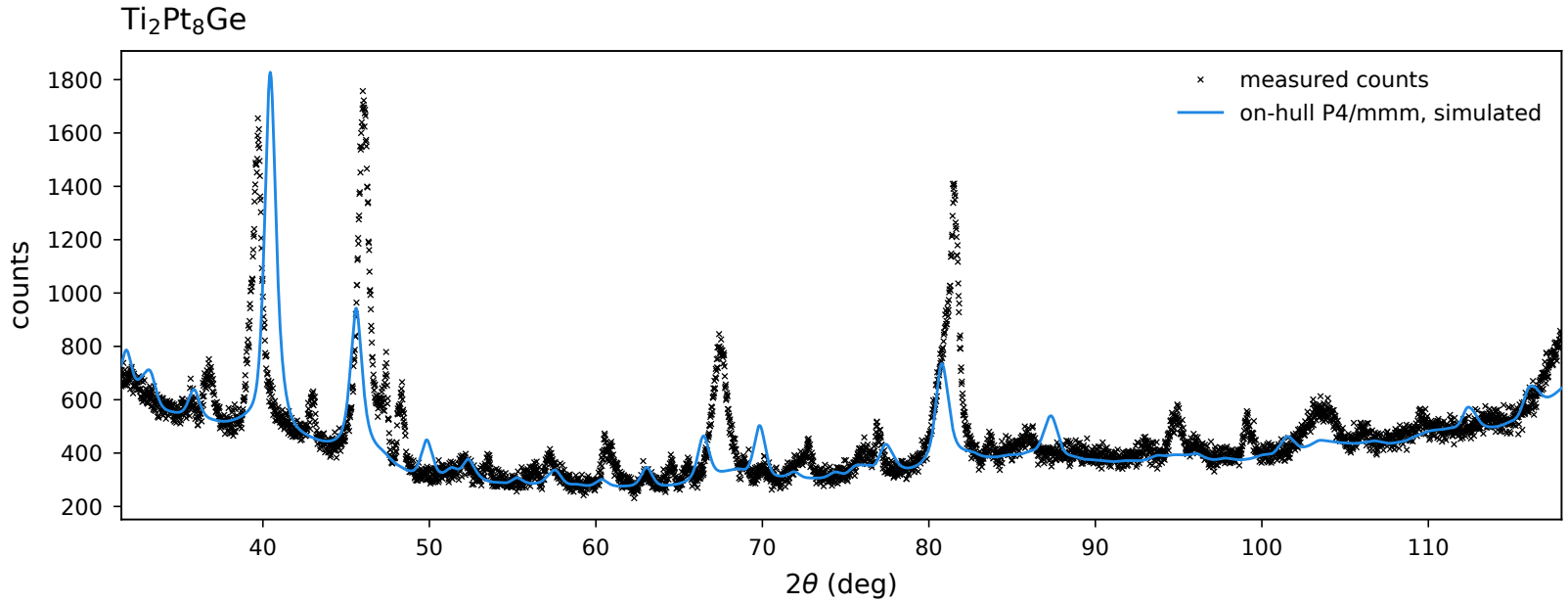

Gd2Fe3Ge5
1 Gd(FeGe)2 25 wt%
2 GdFe0.37Ge2 CeNiSi2 49 wt%
3 Gd0.5Fe3Ge3 YCo6Ge6 24 wt%
4 Ge 2 wt%
Rwp = 9.4 %
predicted Gd2Fe3Ge5, simulated
I − Ibg (arb. units)
1
2
3
4
20
40
60
80
100
2θ (deg)

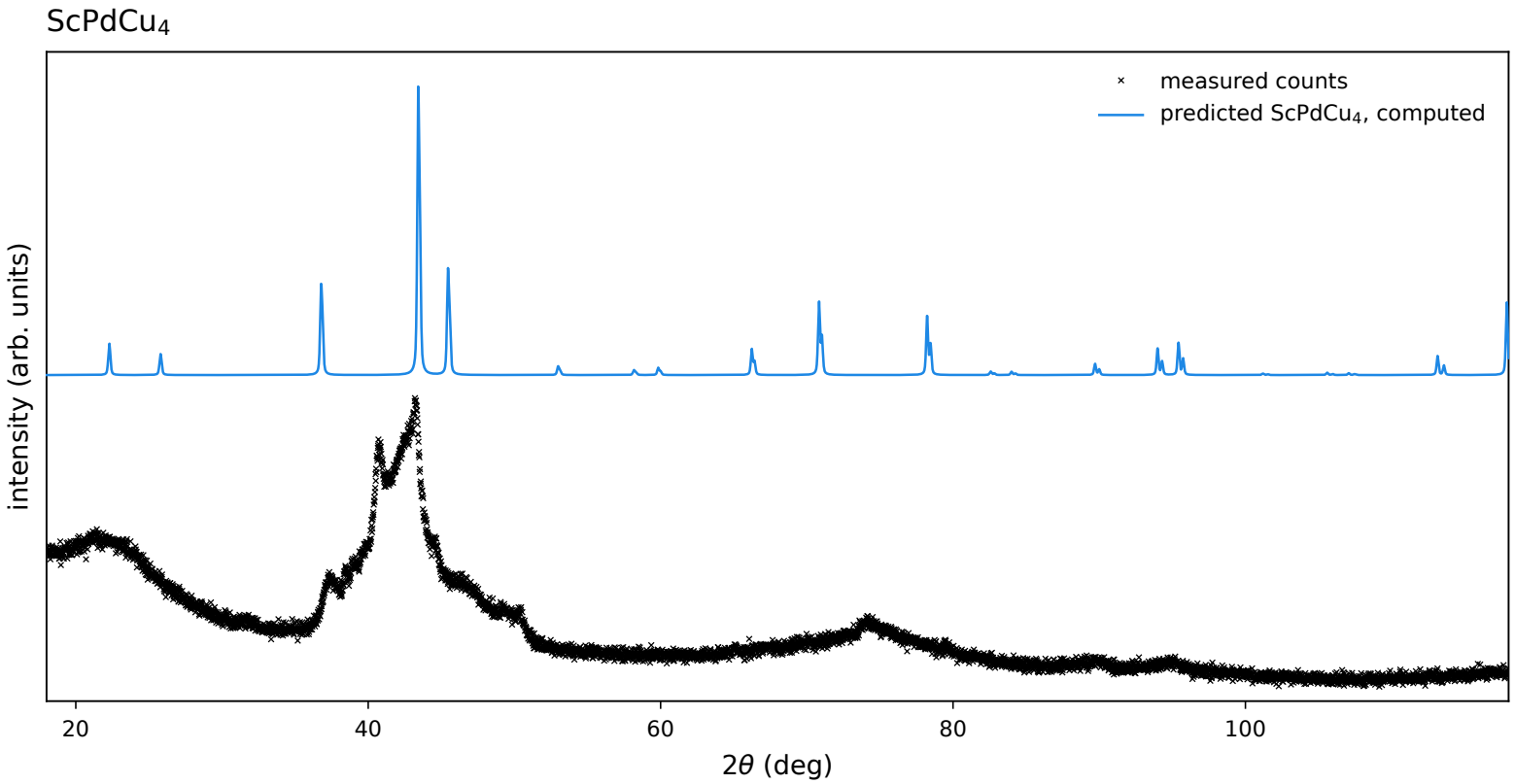

GdFeGe3
1 GdFe0.37Ge2 CeNiSi2 66 wt%
2 Gd0.5Fe3Ge3 YCo6Ge6 21 wt%
3 Ge 13 wt%
Rwp = 7.5 %
predicted GdFeGe3, simulated
I − Ibg (arb. units)
1
2
3
20
40
60
80
100
2θ (deg)

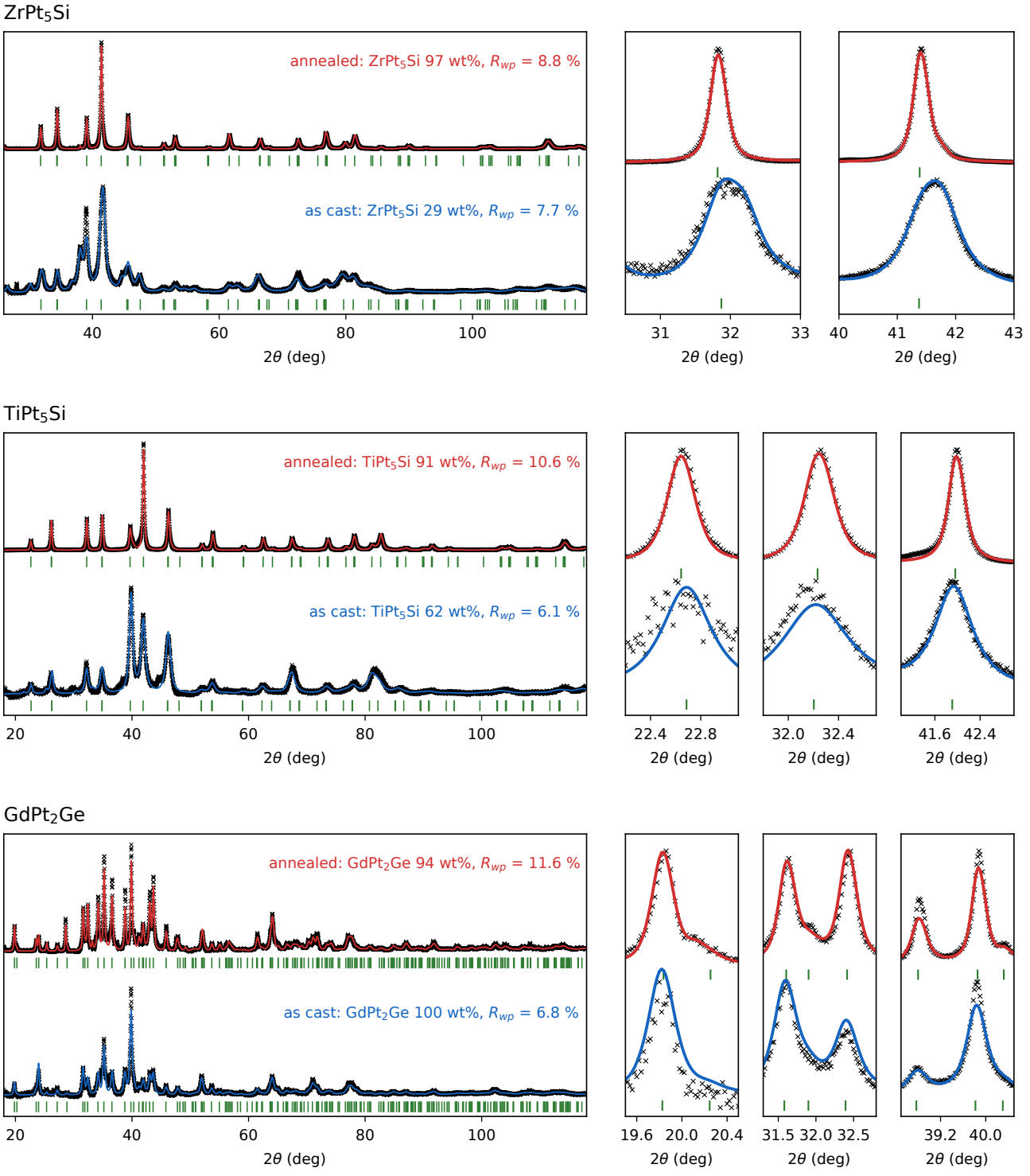

LaRuC2
1 Ru 58 wt%
2 La(OH)3 42 wt%
Rwp = 7.5 %
predicted LaRuC2, simulated
I − Ibg (arb. units)
1
2
20
40
60
80
100
2θ (deg)


Pd6Cu5Ge
measured counts
predicted Cmmm, simulated
counts
2θ (deg)


La2PdAu2
measured counts
predicted La2PdAu2, simulated
counts
2θ (deg)


Ti2Pt8Ge
measured counts
on-hull P4/mmm, simulated
counts
1800
1600
1400
1200
1000
800
600
400
200
40
50
60
70
80
90
100
110
2θ (deg)


ScPdCu4
measured counts
predicted ScPdCu4, computed
intensity (arb. units)
20
40
60
80
100
2θ (deg)


ZrPt5Si
annealed: ZrPt5Si 97 wt%, Rwp = 8.8 %
as cast: ZrPt5Si 29 wt%, Rwp = 7.7 %
40
60
80
100
2θ (deg)
31
32
33
2θ (deg)
40
41
42
43
2θ (deg)
TiPt5Si
annealed: TiPt5Si 91 wt%, Rwp = 10.6 %
as cast: TiPt5Si 62 wt%, Rwp = 6.1 %
20
40
60
80
100
2θ (deg)
22.4
22.8
2θ (deg)
32.0
32.4
2θ (deg)
41.6
42.4
2θ (deg)
GdPt2Ge
annealed: GdPt2Ge 94 wt%, Rwp = 11.6 %
as cast: GdPt2Ge 100 wt%, Rwp = 6.8 %
20
40
60
80
100
2θ (deg)
19.6
20.0
20.4
2θ (deg)
31.5
32.0
32.5
2θ (deg)
39.2
40.0
2θ (deg)